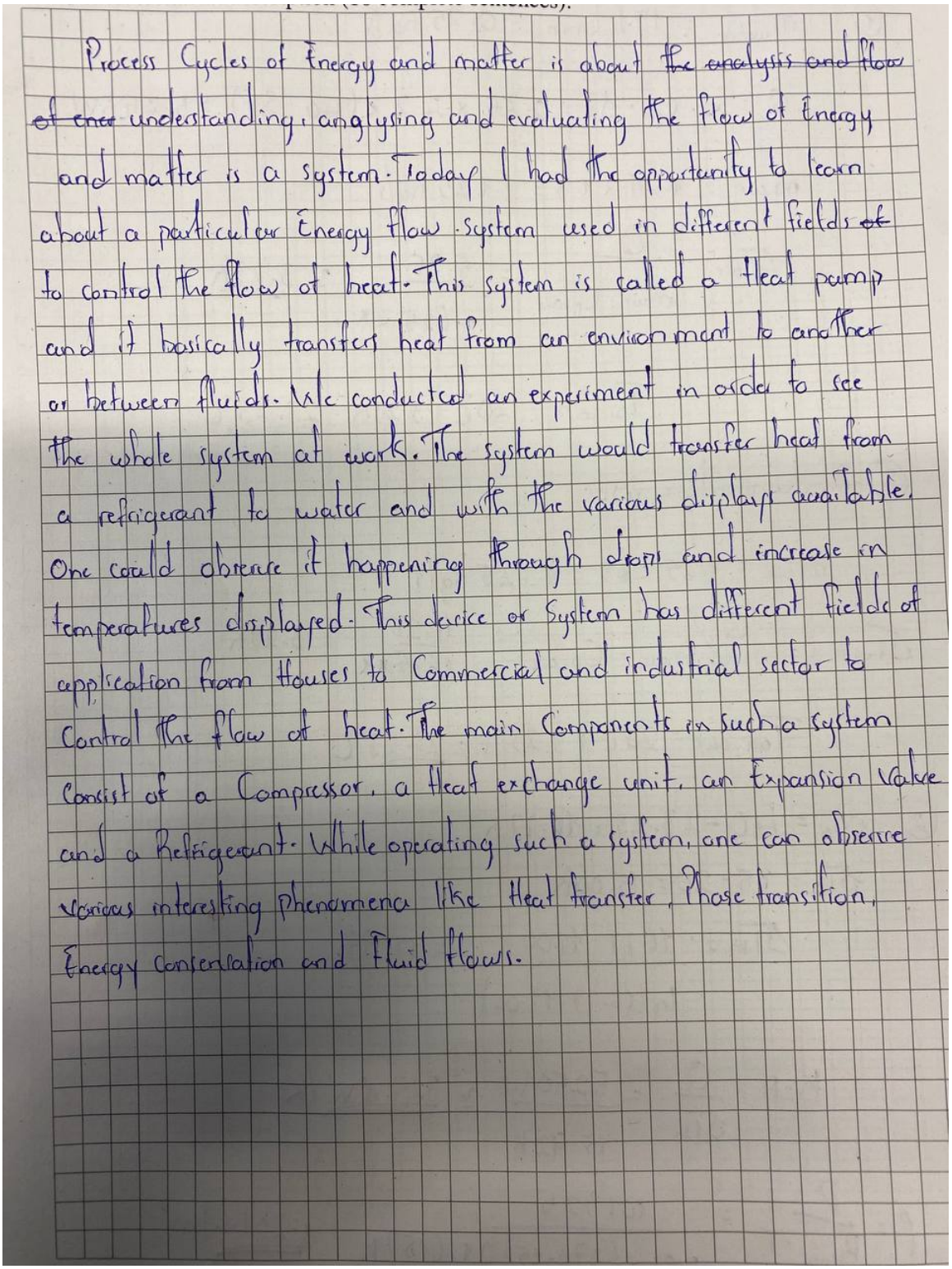
Process Cycle Of Energy and Matter

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Process Cycles of Energy and matter is about ~~the analysis and flow of ener~~ understanding, anglysing and evaluating the flow of Energy and matter is a system. Today I had the opportunity to learn about a particular Energy flow system used in different fields ~~of~~ to control the flow of heat. This system is called a Heat pump and it basically transfers heat from an environment to another or between fluids. We conducted an experiment in orde to see the whole system at work. The system would transfer heat from a refrigerant to water and with the various displays available, One could observe it happening through drops and increase in temperatures displayed. This device or System has different fields of application from Houses to Commercial and industrial sector to Control the flow of heat. The main Components in such a system Consist of a Compressor, a Heat exchange unit, an Expansion Valve and a Refrigerant. While operating such a system, one can observe various interesting phenomena like Heat transfer, Phase transition, Energy Conservation and Fluid flows.

Nice, simplified version of your process flow scheme (block diagram) of process (3-6 blocks):

① $\dot{m}_{water} = 15\,L/min = 0.25\,kg/s$

$\dot{Q} = \dot{m} \cdot c_p \cdot \Delta T = 0.25 \times 4.2 \times (40 - 35) = 5.25\,kW$

$\dot{m}_{air} = 1.25 \times 1.3 = 1.625\,kg/s$

$\Delta T_{air} = \frac{\dot{Q}}{\dot{m}_{air} \cdot c_{p,air}} = \frac{5.25}{1.625 \times 1} \approx 3.23\,K$

$T_{air,out} = 20°C + 3.23K = 23.23°C$

~~② Air temperature rise:~~

~~$\dot{m}_{air} = 1.25 \times 1.3 = 1.625\,kg/s$~~

~~$\Delta T_{air} = \frac{\dot{Q}}{\dot{m}_{air} \cdot c_{p,air}} = \frac{5.25}{1.625 \times 1} = 3.23K$~~

~~$T_{air,out} = 20°C + 3.23K = 23.23°C$~~

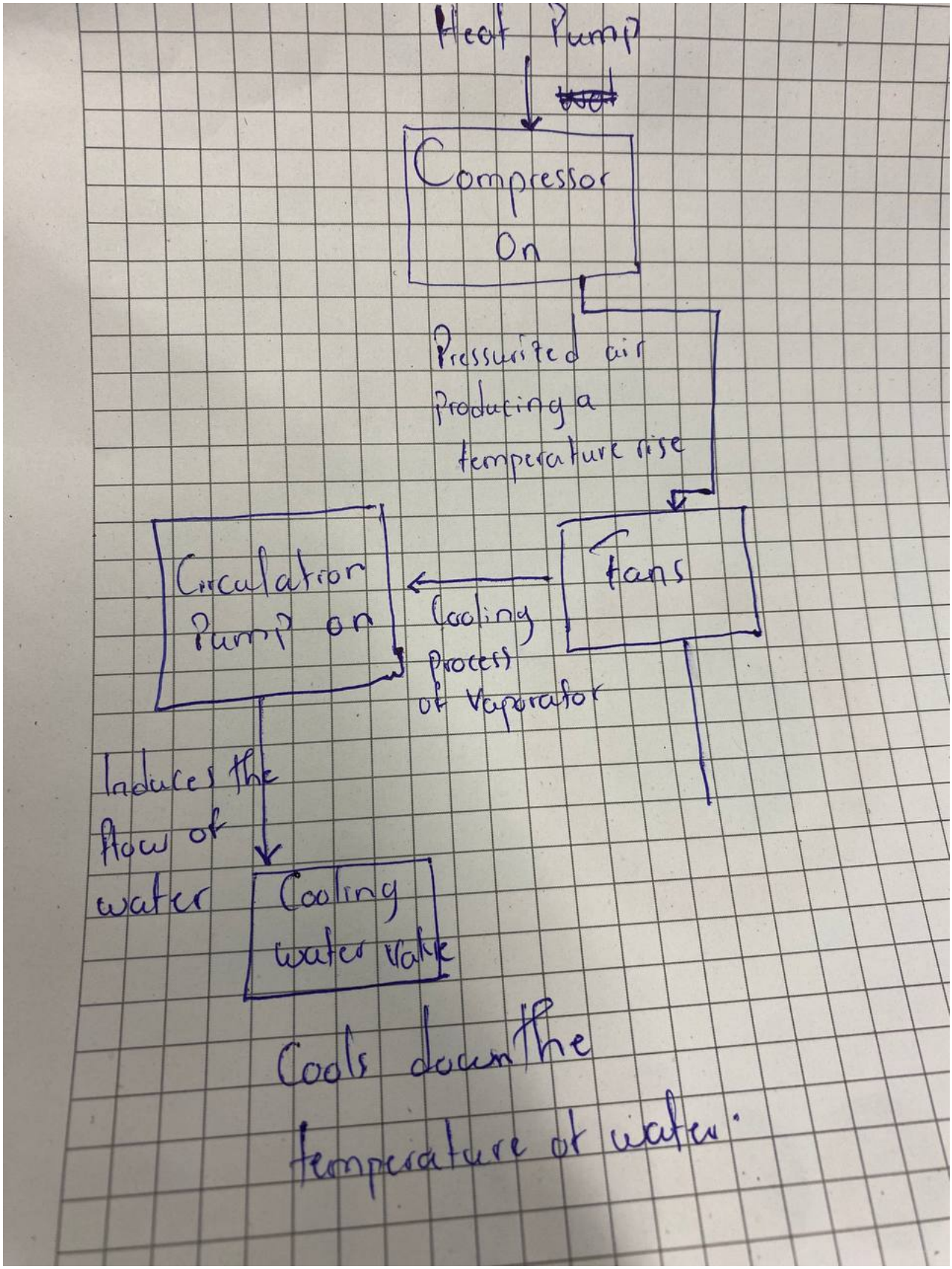
② $\Delta T_1 = 40 - 23.23 = 16.77K$

$\Delta T_2 = 35 - 20 = 15.00K$

$\Delta T_{ln} = \frac{16.77 - 15.00}{\ln(16.77/15.00)} \approx 15.9K$

$K \cdot A = \frac{\dot{Q}}{\Delta T_{ln}} = \frac{5250\,W}{15.92\,K} \approx 329.8\,W/K$

③ $\rho = \frac{p}{R_{spec} \cdot T} = \frac{101325}{287 \times (273.15 + 21.615)K} = 1.20\,kg/m^3$


Heat Pump
Compressor On
Pressurised air producing a temperature rise
Fans
Cooling process of Vaporator
Circulation Pump on
Induces the flow of water
Cooling water valve
Cools down the temperature of water.